4. ESTIMATION OF PROTEIN BY LOWRY'S METHOD

Aim:

To estimate the amount of total protein (BSA) in the given test solution.

Principle:

It is the most commonly used method for determination of proteins in cell free extracts because of its high sensitivity and quantities as low as 20µg of proteins can be measured. The CO-NH (peptide bonds) in polypeptide chain reacts with copper sulphate in an alkaline medium to give a blue coloured complex. In addition, tyrosine and tryptophan residues of proteins cause reduction of the phosphomolybdate and phosphotungstate components of the Folin-Ciocalteau reagent to give bluish products which contribute towards enhancing the sensitivity of this method

Reagents:

1. 5% TCA
2. 0.1N sodium hydroxide
3. **Lowry's reagent:** Solution A: 2.0% sodium carbonate in 0.1N sodium hydroxide; Solution B: 0.5% copper sulphate in 1% sodium potassium tartarate solution (freshly prepared); Mix 50ml of solution A and 1ml of solution B prior to use.
4. **Folin'sciocalteu reagent:** A commercial preparation is diluted with water in the ratio of 1:2 before use.
5. **Stock solution:** 100mg BSA is weighed and made upto 100ml of distilled water
6. **Working standard solution:** 10ml of the stock is diluted to 100ml using distilled water

   Concentration = 100 µg / ml

Procedure:

0.5-2.5ml of working standard BSA solution is pipetted out into five different test tubes and labelled as S1-S5 with the concentration of 50-250µg. 0.5ml of the test solution is pipetted out into a test tube and labelled as T_1. Final volume is made upto 2.5 ml using 0.1N sodium hydroxide in all the tubes. A blank is also prepared simultaneously comprising 2.5ml of 0.1N sodium hydroxide. 4.5ml of lowry's reagent is added to all the test tubes followed by the addition of 0.5ml of Folin's reagent. All the test tubes are kept at room temperature for 20 minutes. The colour developed is measured at 660nm using a colorimeter.

A standard graph is drawn with optical density in Y- axis vs. concentration of BSA in X axis. The amount of BSA present in the given test solution is then calculated.

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